FINITE TYPE DESCENDS ALONG A FAITHFULLY FLAT RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Expansion of an element of a base change as a finite sum. Used in the counted proof at lines 31671-31674. Produces the finitely many elements of the source that generate after base change.
- Faithful flatness as detection of equality of submodules. Used at the end of the counted proof (lines 31675-31677). Turns equality of the extended subalgebras back into equality in the source.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $R \to R'$ be a faithfully flat ring map. Set $S' = R' \otimes_R S$. Then $R \to S$ is of finite type if and only if $R' \to S'$ is of finite type.*

Proof. It is clear that if $R \to S$ is of finite type then $R' \to S'$ is of finite type. Assume that $R' \to S'$ is of finite type. Say $y_1, \ldots, y_m$ generate S' over R'. Write $y_j = \sum_i a_{ij} \otimes x_{ji}$ for some $a_{ij} \in R'$ and $x_{ji} \in S$. Let $A \subset S$ be the R-subalgebra generated by the x_{ij}. By flatness we have $A' := R' \otimes_R A \subset S'$, and by construction $y_j \in A'$. Hence $A' = S'$. By faithful flatness $A = S$. □

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